FH Vorarlberg
University of Applied Sciences

Energy and Storage System - Economics

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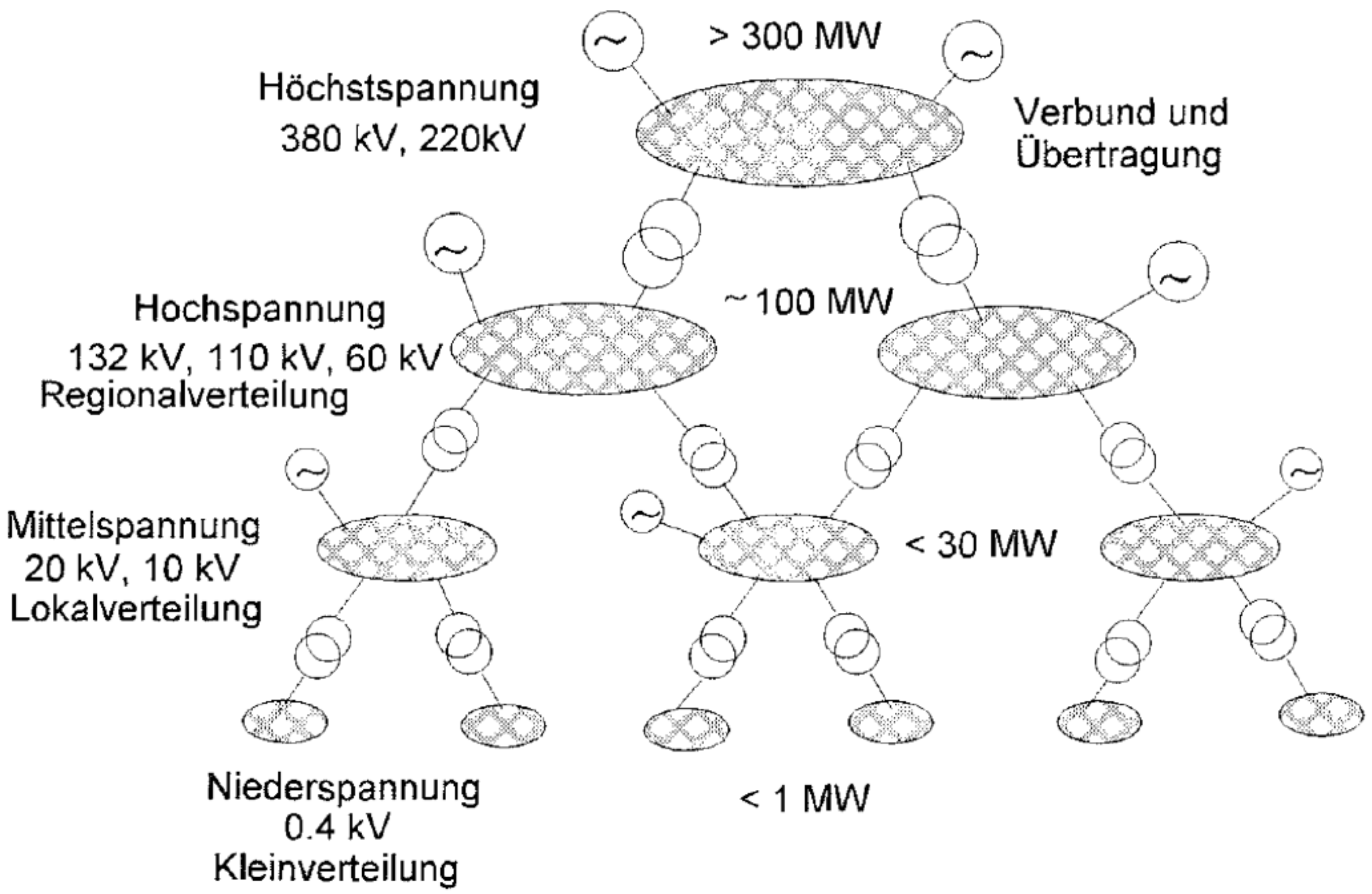
Course Number 47083

Primary Energy - Production

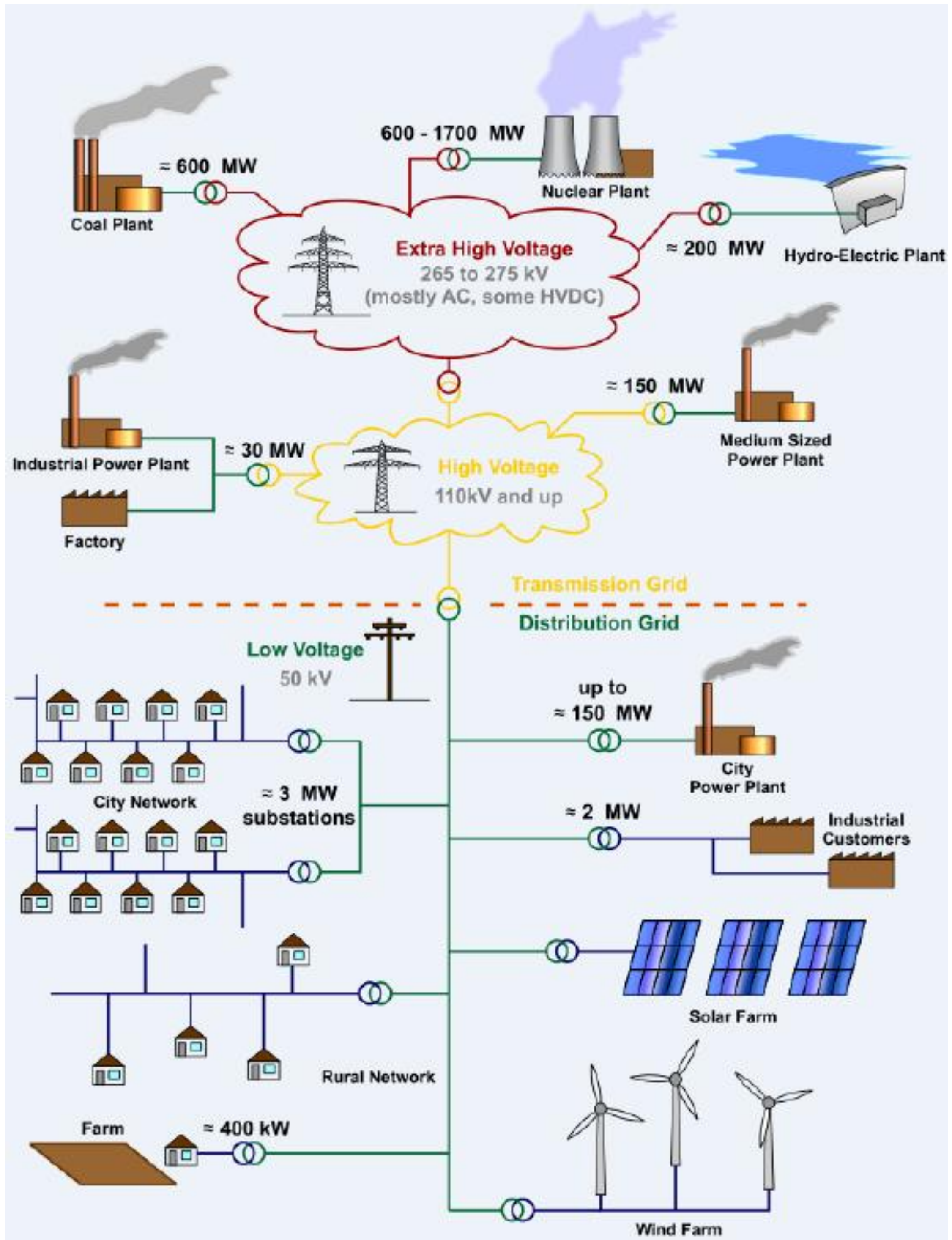
- Primary Energy transformed via power generators into electricity (Secondary Energy)
- High power producers (100 MW) connected t High Voltage transmission lines
- Low power producers (30 MW) are connecte to Medium voltage transmission lines



Primary Energy - Production

- High voltage lines carrying high amount of electrical power

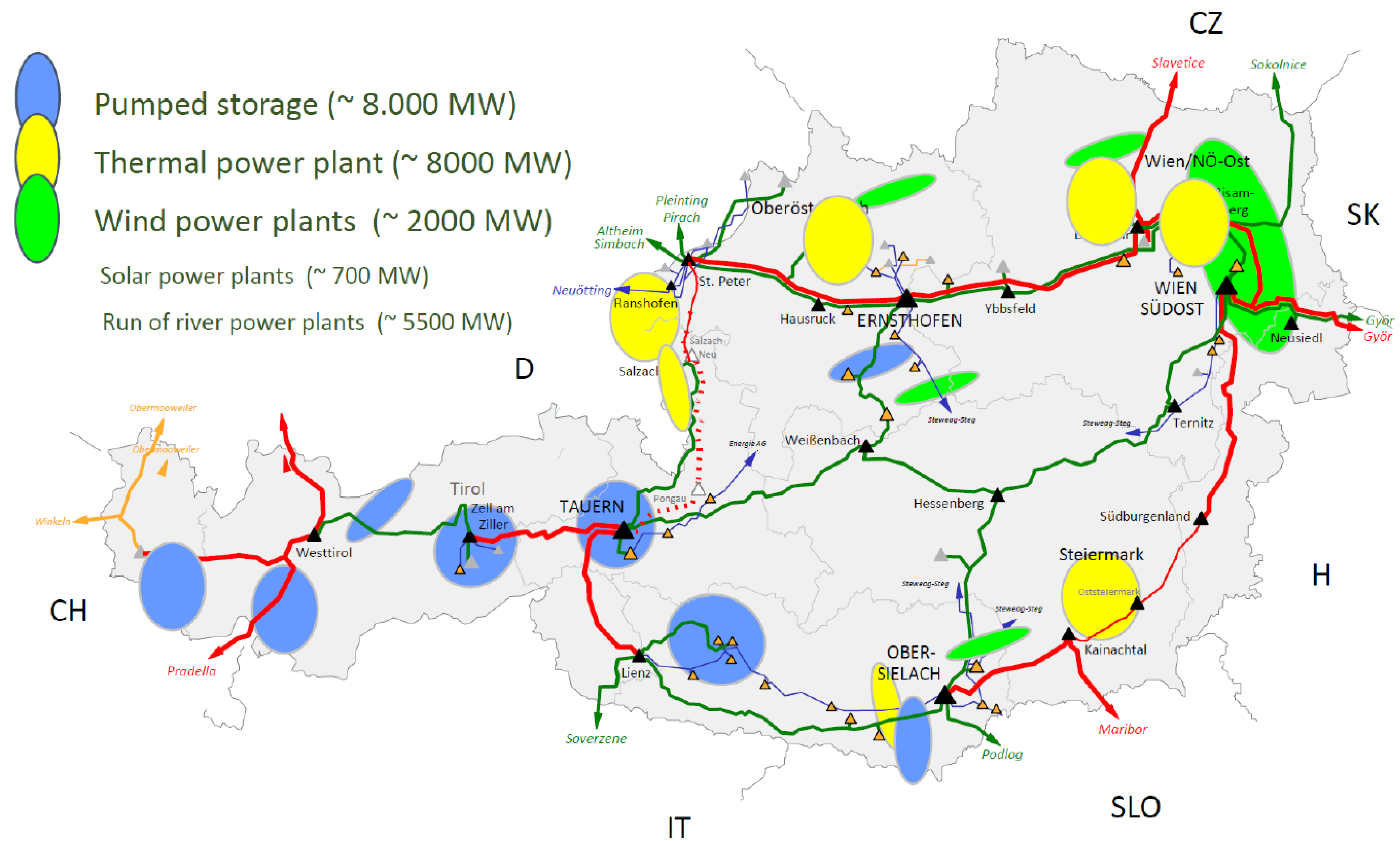
- From the production place to consumer hubs



Energy Production - Production

- Power plant connections in Austria

- EU project Salzburg – 380 kV ring has been closed

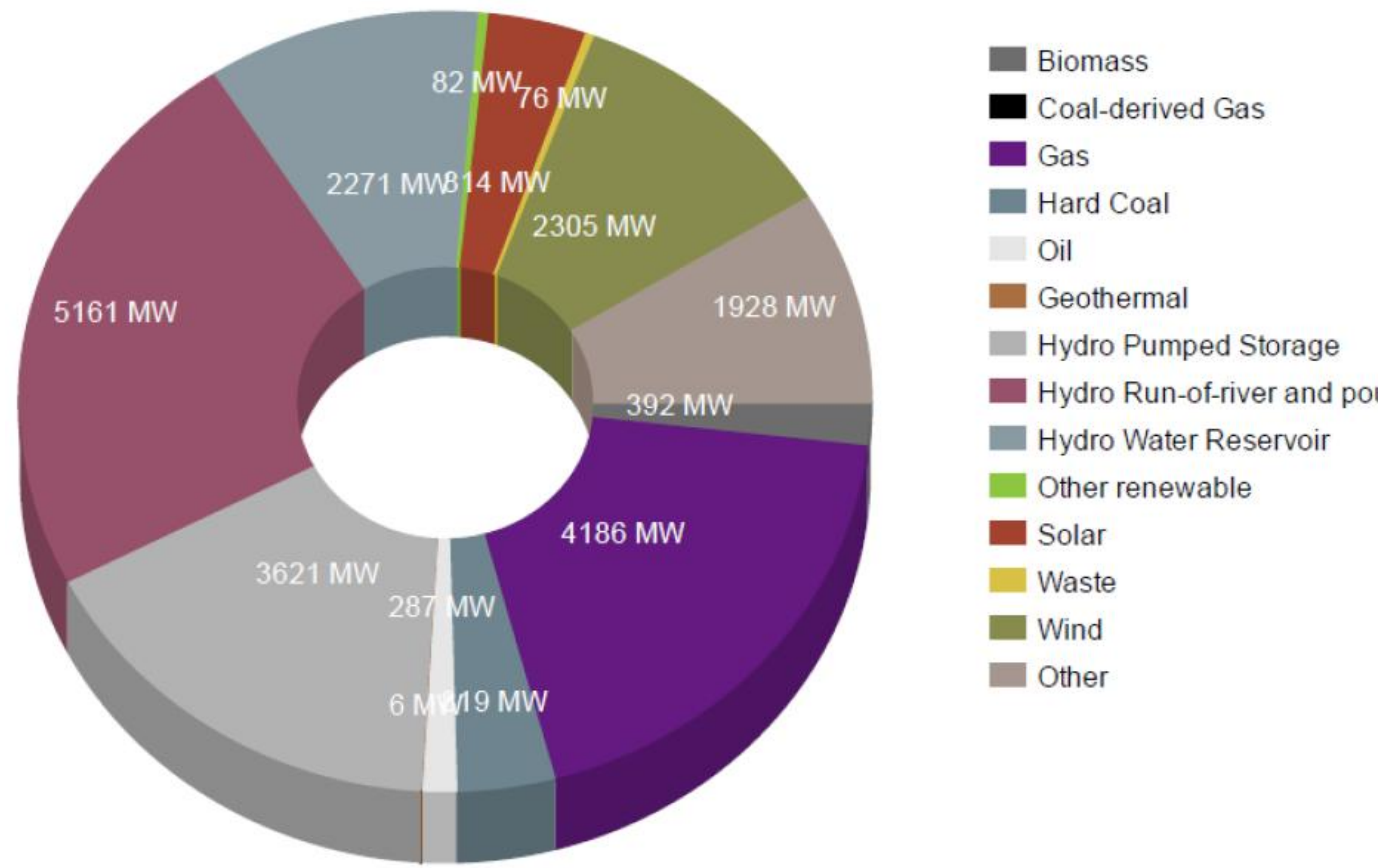


Energy Production - Austria

- Installed capacity in Austria with reference to primary resources

Year: 2016

Type of power plant	Sum of installed net capacity of the respective type of power plant in MW
Biomass	392.769
Coal-derived Gas	0.000
Gas	4,186.579
Hard Coal	819.000
Oil	287.722
Geothermal	6.410
Hydro Pumped Storage	3,621.840
Hydro Run-of-river and poundage	5,161.037
Hydro Water Reservoir	2,271.380
Other renewable	82.610
Solar	814.422
Waste	76.449
Wind	2,305.694
Other	1,928.978
Total	**21,954.890**



Structure of electrical energy market

- Level 1 - Production
- Level 2
  - Commercial
  - Power flow
- Level 3 – Consumption

- Former monopolistic companies are split up
  - Generation
  - Transmission
  - Distribution

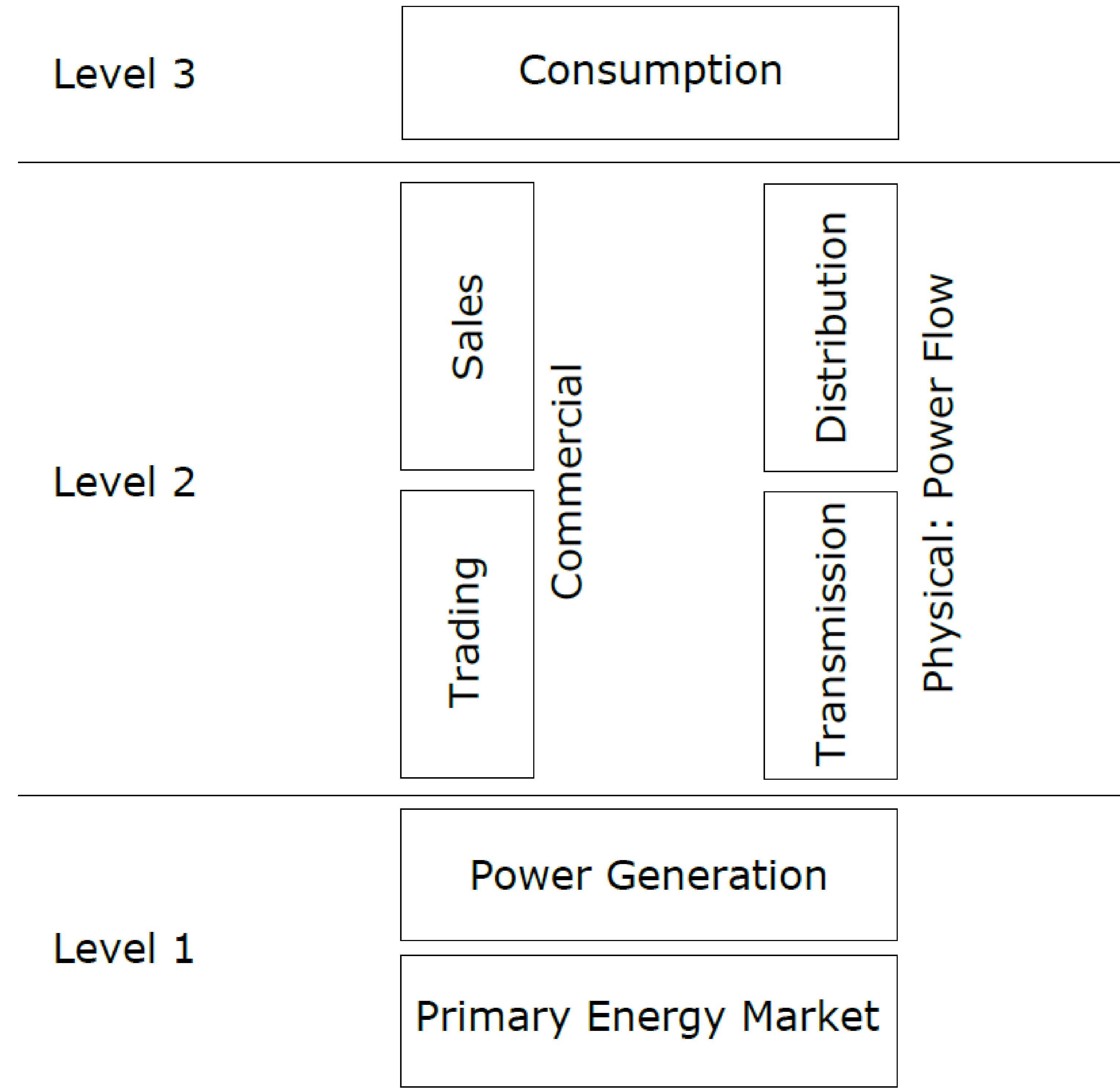


Figure 18: Structure of the electrical energy market
Source: Moser (2011)

Energy Costs

- Load
  - Base load - Continuous
  - Variable (peak) load – Changes over time

- Base load covered by most efficient plant

- In addition reserve is needed
  - Generator unit trips because of a fault
  - Unexpected load change
  - Maintenance work
  - Error in load prediction
  - Compensating fluctuating power producer (E.g. wind)

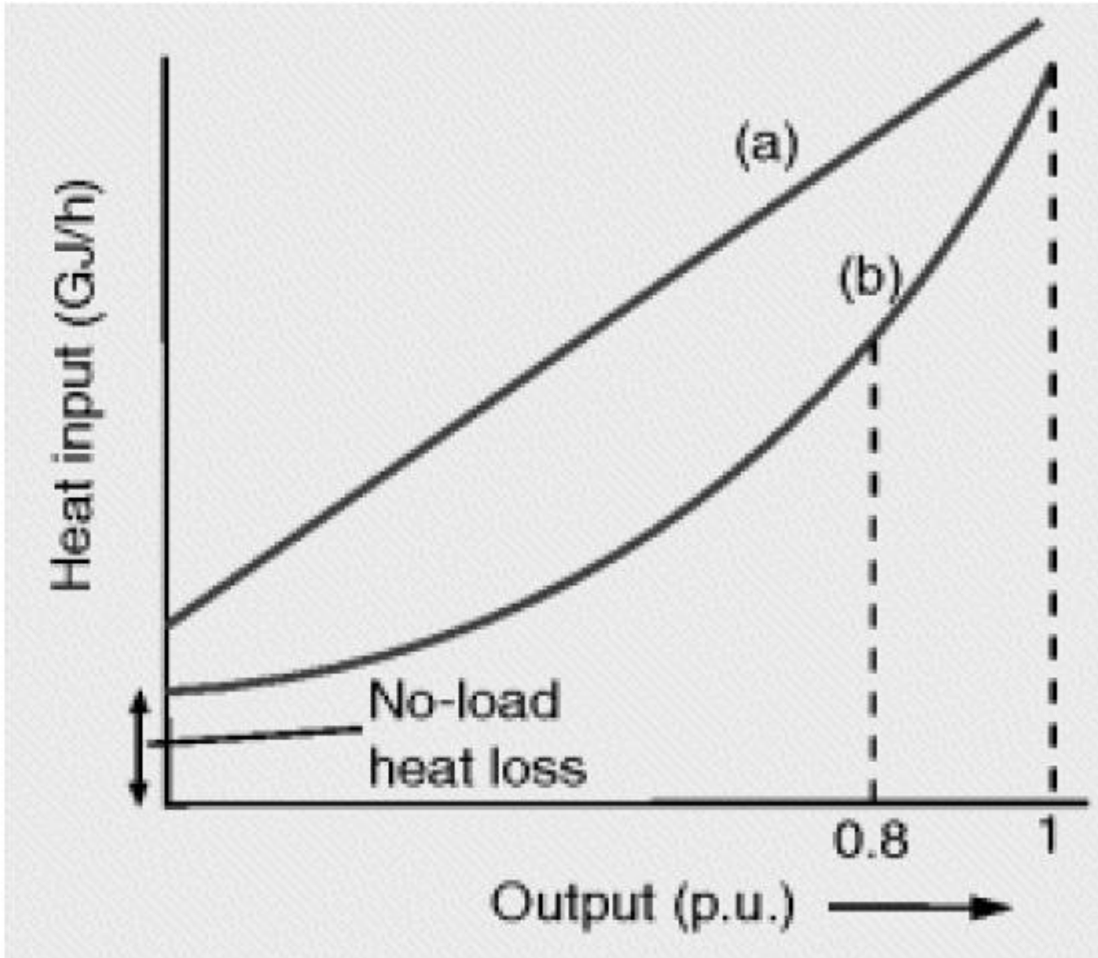
Energy Costs

- Proportion of generator power kept as reserve
  - E.g. Generator being operated at 75% of their full capability
  - Called spinning reserve
- Usually a mix of different power plants
  - Hydro, coal, renewables
  - Different operation costs at different operation points
- Optimum mix = most economic operation

Generating Technology	Capital Cost of Plant £/MW	Cost of electricity £/MWh
Combined Cycle Gas Turbine	720	80
Coal	1800	105
Onshore wind	1520	94
Nuclear	2910	99
Data from UK Electricity Generating Costs Update, 2010, Mott MacDonald, reproduced with permission		

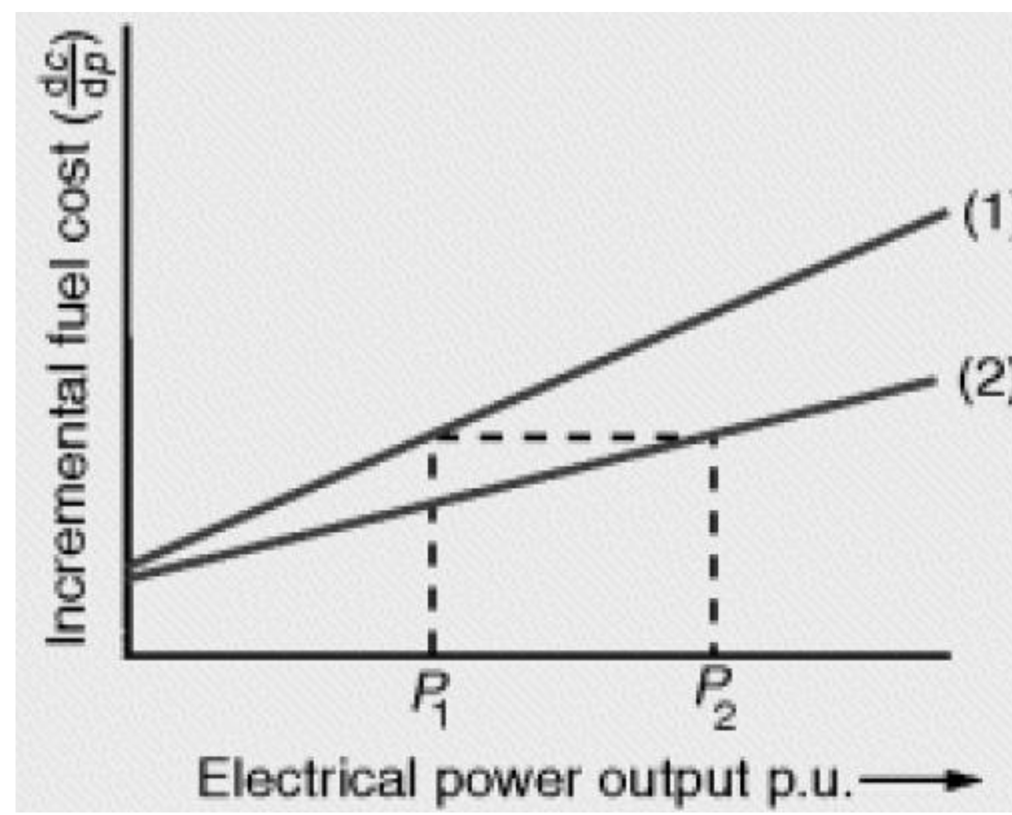
Energy Costs for thermal power plants

- Wilans line
  - Heat rate input [GJ/h] vs. power output [MW]
  - $H_{(P)} = a + bP + cP^2$



Energy Costs

- Find optimum generator unit for a specific allocated load
  - Incremental fuel cost [Euro/MWh] vs power output [MW]
  - $C_{(P)} = H_{(P)} x$ Cost of Energy



Structure of electrical energy market

- Goal of unbundling
  - More competition
  - More efficiency
- Electrical energy becomes a regular tradable **product**
  - Market for electrical energy
  - Purchaser and Seller are agreeing on a price

- However
  - Electrical energy cannot be stored in larger quantities
  - Power flow follows physical rules

Structure of electrical energy market

- There is only one interconnected electrical infrastructure
  - Makes transmission and distribution an monopolistic activity

- Because of that
  - Transmission and distribution are still regulated (by E-control in Austria)

  - The product “electrical energy” is time based

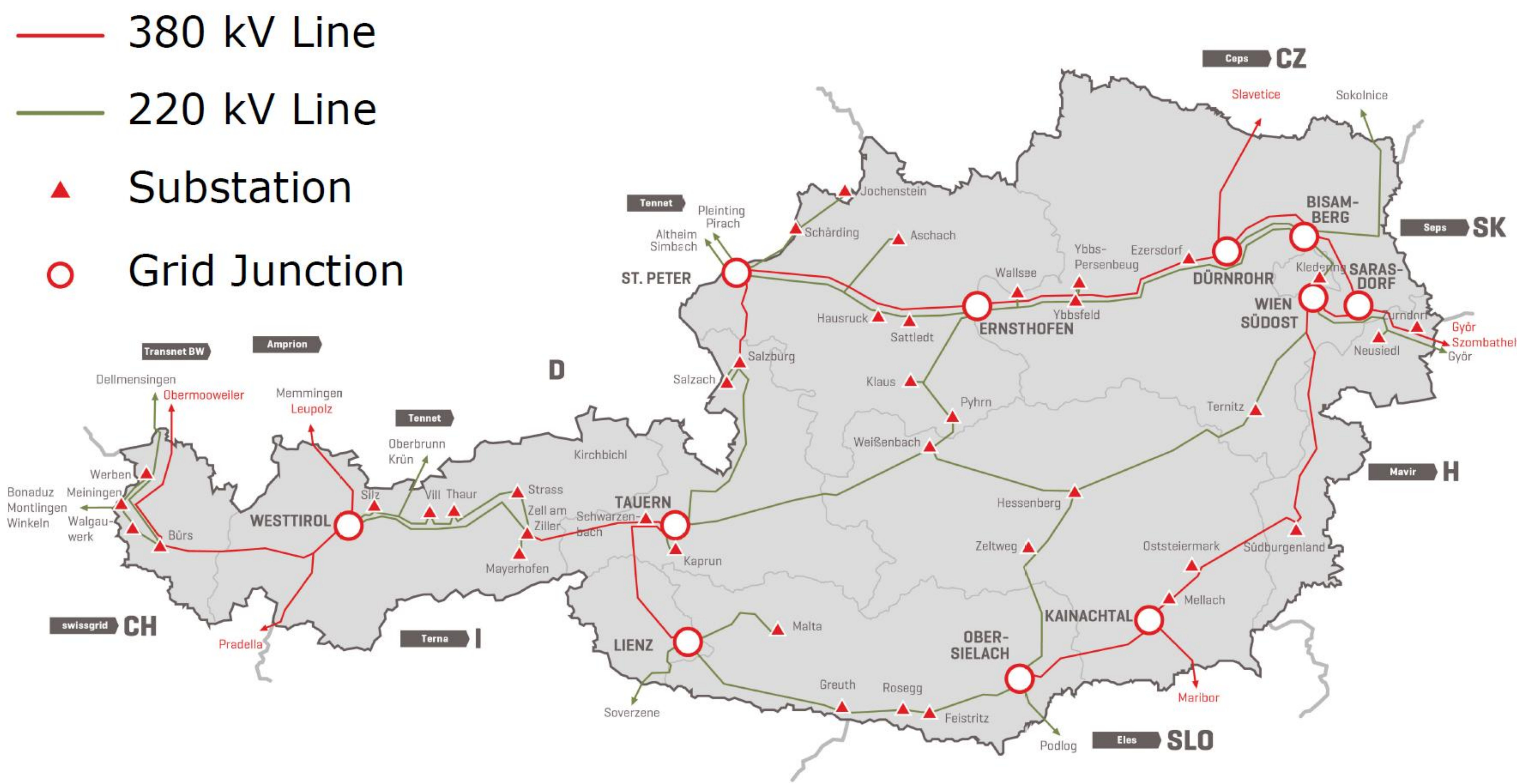
Structure of electrical energy market

- Managing of the physical interconnection system
  - Control area manager CAM (dt. Regelzonenführer RZF)

- Managing includes
  - Keep power balance
  - Keep constant frequency
  - Keep operation within rigid security limits
  - Maintain a safe and reliable system
  - Assuring adequate voltage support

- CAM: Austrian Power Grid APG

Austrian High Voltage Grid



Structure of electrical energy market

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- CAM for Austria: Since 2012 - Austria Power Grid APG (prev. TIWAG and VKW

Control areas in Germany, Switzerland

- Germany
  - Amprion (prev. RWE Transportnetz)
  - 50 Hertz Transmission (prev. Vattenfall)
  - TransBW (prev. EnBW Transportnetze)
  - TenneT TSO (prev. E.ON Netz)

- Switzerland
  - Since 2009 – Swissgrid (prev. up to 8 control areas)

Power Generation Costs

- CAPEX
  - Capital Expenditures
  - Dt. Investitionskosten

- OPEX
  - Operational Expenditures
  - Dt. Betriebskosten

Power Generation Costs - CAPEX

- Funds for
  - Acquire or upgrade major physical assets (dt. Anlagengüter)
  - Assets are used to generate profit

- Includes
  - Repairing costs
  - Improvements to extend productive life
  - Build a new plant

Power Generation Costs - CAPEX

- Deprecation (dt. Abschreibung)
  - Assets useful life extend more then one year
  - Costs of asset are spread over its designated life
  - Determined by tax regulations

- Compared to other industries
  - High investment costs for major physical assets
  - That includes
    - Designing, approvals
    - Construction
    - Asset costs and commissioning of an plant

Power Generation Costs - OPEX

- Operational Expenditures OPEX
  - Funds covering the ongoing costs
  - Used to run basic business

- Can be divided into
  - Production dependent operation costs
    - Fuel costs
    - Emission rights (for fossil fuels)
  - Other operation costs
    - Costs for labor
    - Administration
    - Maintenance work

Power Generation Costs

- Annual costs C for electrical power is the sum of the investment costs C_I (CAPEX) and the operational costs C_O (OPEX)

$$C = C_I + C_O$$

Investment Costs

- Annual investment costs C_I [€ / year] consists of the specific investment costs k_I [€/kW], the annuity p [%/year] (dt. Tilgungsrate) and the rated power output P [W]

$$C_I = k_I p P \cdot 10$$

Specific Investment Costs k_I [€/kW]

- Technical and economical parameter of reference power plants put into operation 2010

Source	Plant Type	Power	Eff.	Life Time	k_I	Load f_L
		[MW]	%	[a]	[Euro/kW]	%
Gas	CCGT	1000	60	30	480	85
Hard Coal	Steam PP	1020	46	35	1000	85
Lignite Coal	Steam PP	1050	44	35	1150	85
Uranium	PWR	1600	36	60	1850	85
Hydro	River PP	3.1		60	4982	57
Wind	WT onshore	2		20	1050	15-25
Wind	WT offshore	5		20	1950	25-40
Solar	PV Plant	0.5		25	4275	10
Solar	PV Roof	0.002		25	5200	10
Bio mass	Wood	20	35	20	2100	85

Operational Costs

- The operational costs C_O [€/year] are defined as fuel factor k_F [€/h] (dt. Brennstoff-faktor) multiplied by the effective operating time tau_{eff} (Dt. effective Betriebszeit) [h/year]

$$C_O = k_F \tau_{eff}$$

Fuel factor k_F

- The fuel factor k_F [€/h] gets calculated by multiplying the specific heat consumption *3600/eta* [kJ/kWh], the fuel cost *f* [€/GJ], the fuel handling costs f_F
- Fuel handling costs
  - fuel preparation
  - disposal of the fuel residues (dt. Rückstände)

$$k_F = \frac{3600}{\eta} f_F f P \cdot 10^{-3}$$

Effective Operating Time

- The effective operating time *tau*$_{eff}$ [h/year] is based on
  - Utilization of available power f_L (dt. Auslastung)
  - Availability f_A (dt. Verfügbarkeit) of the generation unit itself

$$\tau_{eff} = f_L f_A \cdot 8760$$

Utilization of power

- The availability f_A (dt. Verfügbarkeit) is defined as the actually produced electrical energy W_L a year [MWh/a] divided by the amount of energy which would have been produced W_A [MWh/a]
- Actually produced energy W_L = rated Power output P [MW] multiplied by the time the machine was operating $t_{Operating}$ [h]

$$f_L = \frac{W_L}{W_A}$$

Utilization of power

- Possible energy W_A = rated Power output P [MW] multiplied by the time the machine was ready to be operated, the operational readiness t_{Ready} [h]
- Availability is defined as the operational readiness (Dt. Betriebsbereitschaft) per year (8760 hours)

$$f_A = \frac{t_{Ready}}{8760h/a}$$

Annual Electricity Production

- Annual electricity production $E_{Produced}$ (Dt. Jahresstromabgabe) [kWh/year] is calculated by multiplying the rated power output [W] by the effective operating time tau_{eff}

$$E_{Produced} = P\tau_{eff} \cdot 10^3$$

Power Generation Costs – Calculation

- The power generation costs *epsilon* [€/kWh] are the annual costs C for electrical power generation divided by the annually produced electricity $E_{Produced}$

$$\epsilon = \frac{C}{E_{Produced}}$$

Generation System Planning

- The effective operating time tau_{eff} influences the power generation costs significantly

- Goal of generation planning
  - consider individual generation technology
  - define minimum operation hours for each type

Generation System Planning

- Define load situation in order to choose the right power generation technology
- Daily load curves are accumulated

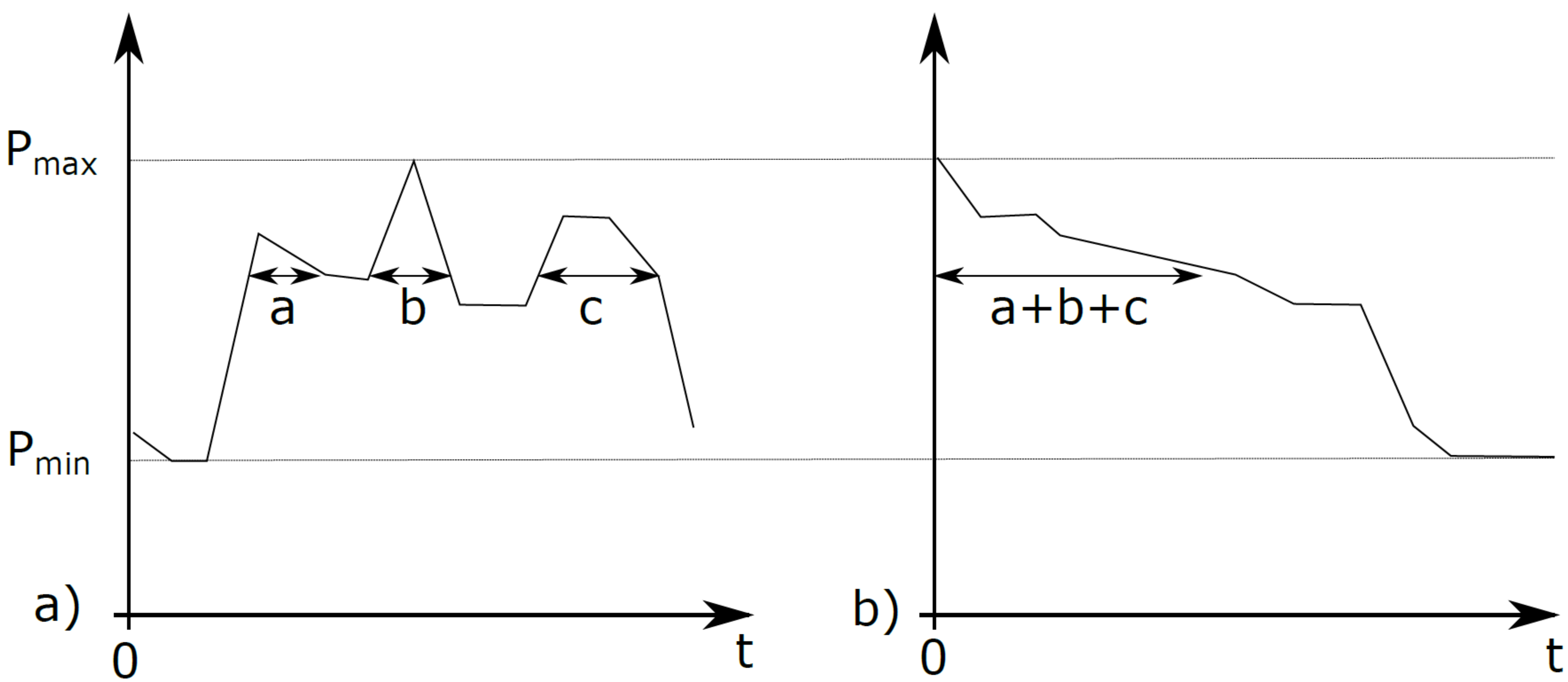


Figure 20: Daily load curve (a) and accumulated daily load curve (b)
Source: Moser (2011)

Classification of operating hours

- t_2: Peak load (dt. Spitzenlast)
- t_1: Mid-merit load (dt. Mittellast)
- t_0: Base-load (dt. Grundlast)

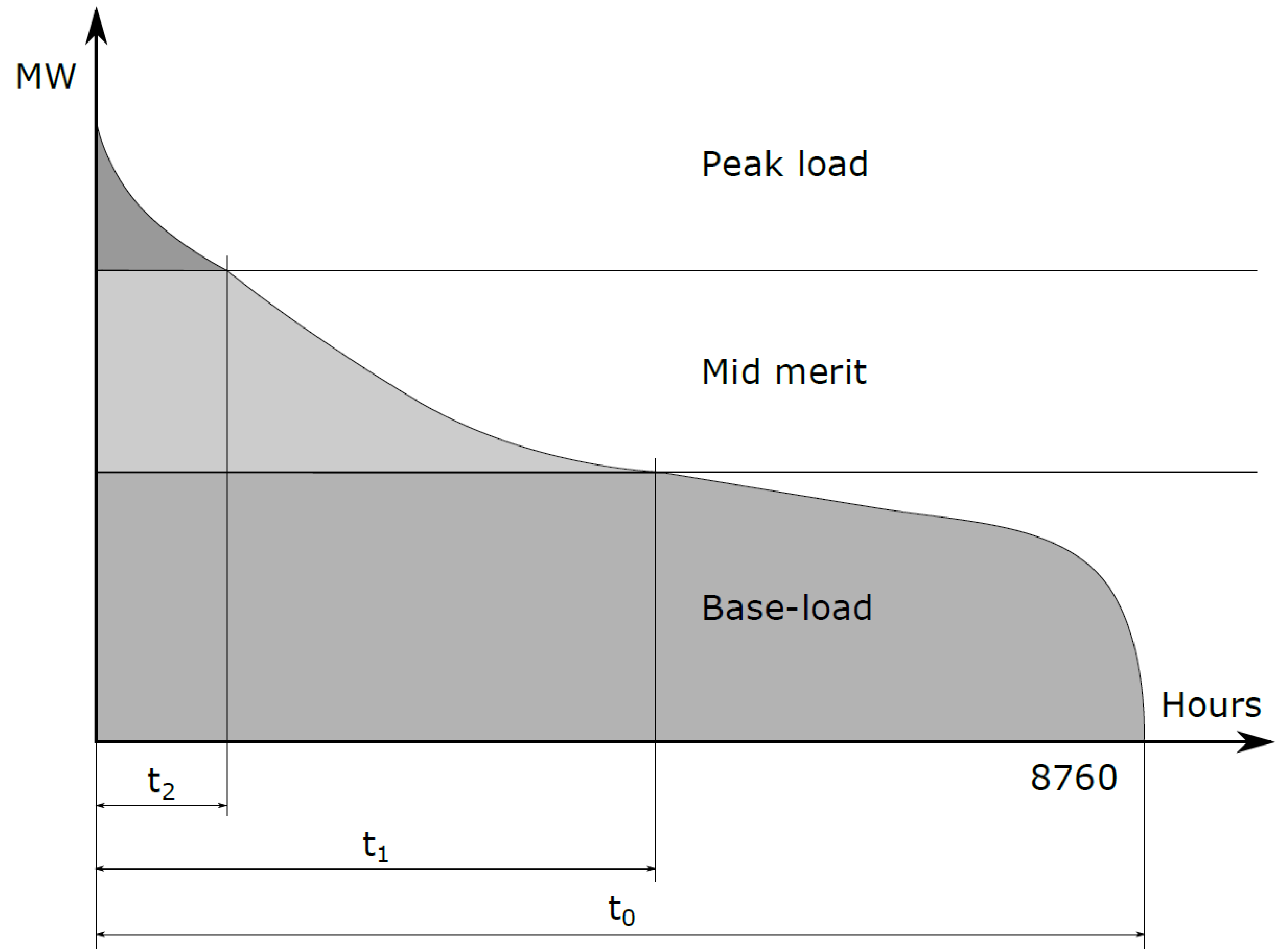


Figure 21: Accumulated annual load duration curve

Base Load

- Constant electrical power during the entire day

- Power plant types used
  - Nuclear power plants
  - Coal power plants (Lignite)
  - Hydro power plants (River)

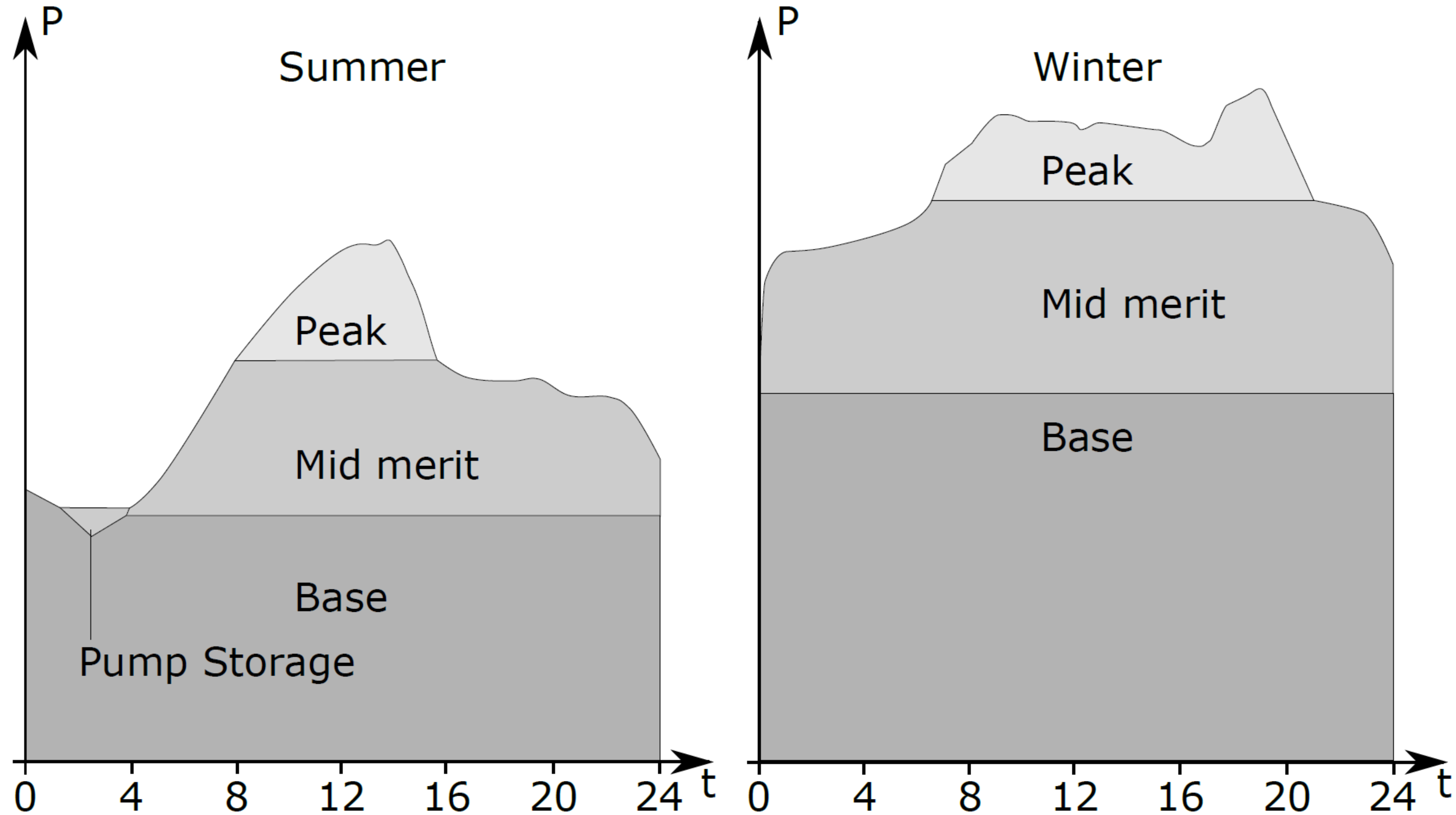
Peak Load

- During lunch time and in the early evening the load curve shows high incrase in power consumption
- Relatively short period of time (few hours)
- Characteristic of power plants covering peak loads
  - Start up fast
  - Reach rated power output within minutes

- Power plant types used
  - Pump storage power plants
  - Gas turbine power plants

Mid-Merit

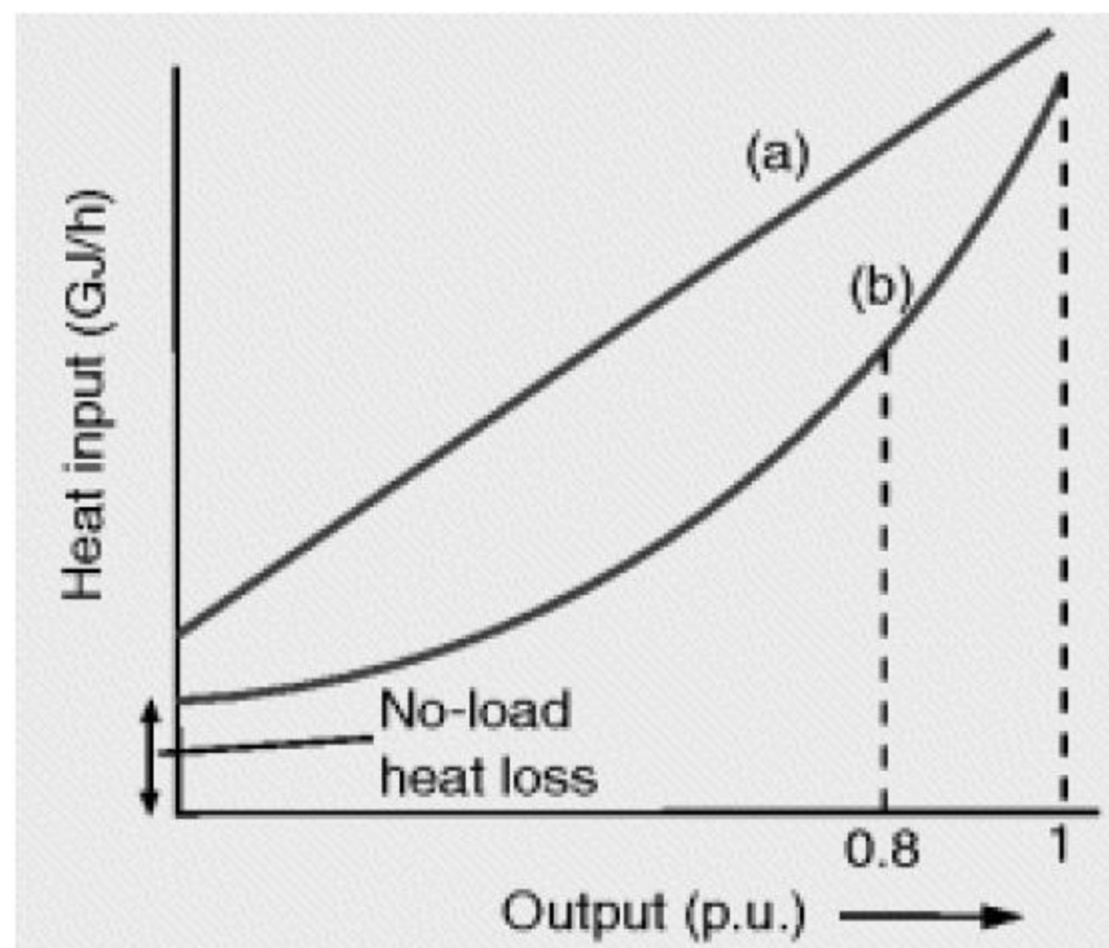
- Between continuous base load and short term peak load
- Cover for the increased power consumption during a day
- At night
  - Switched off
  - Significantly reduced output
- Power plant type used
  - Coal power plant fired with hard coal



Energy Costs – Calculation Example

Assignment 1:

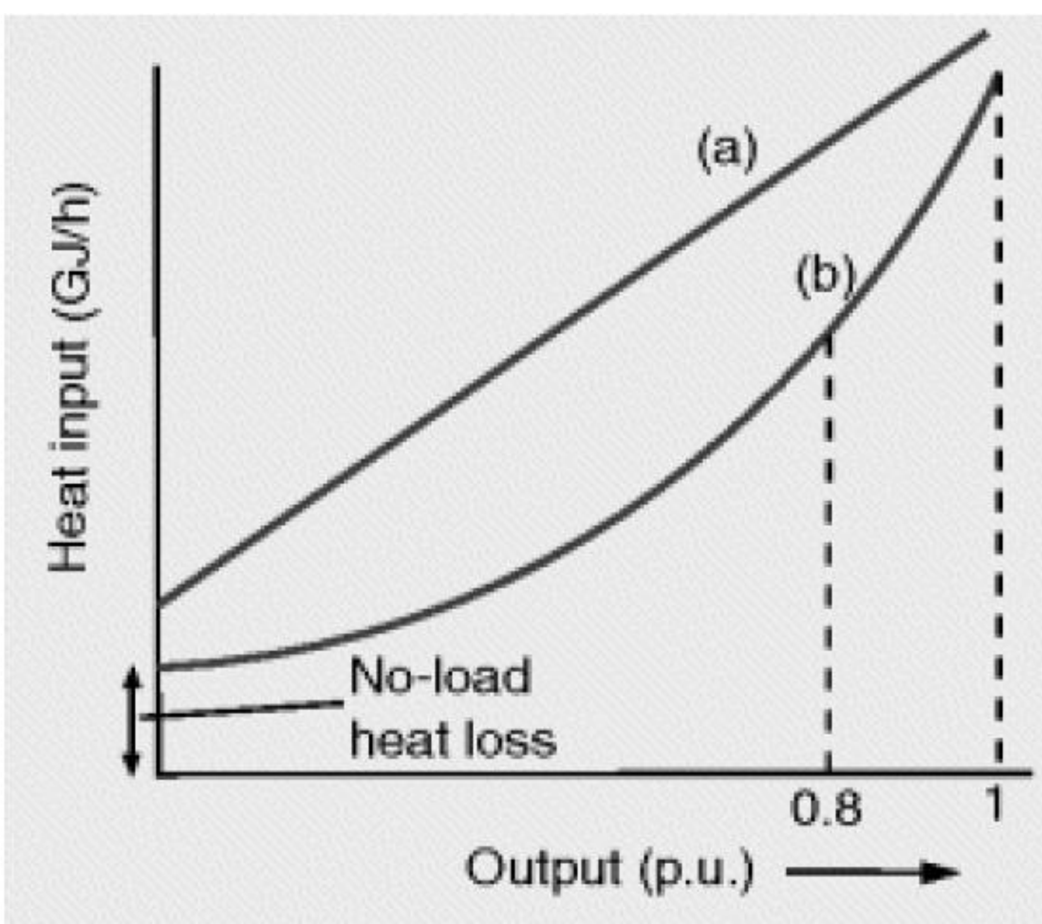
- What is the required heat input for a coal power plant to produce 500 MW electrical energy.
- Wilans line: $H_{(P)} = 126 + 8.9P + 0.0029P^2$

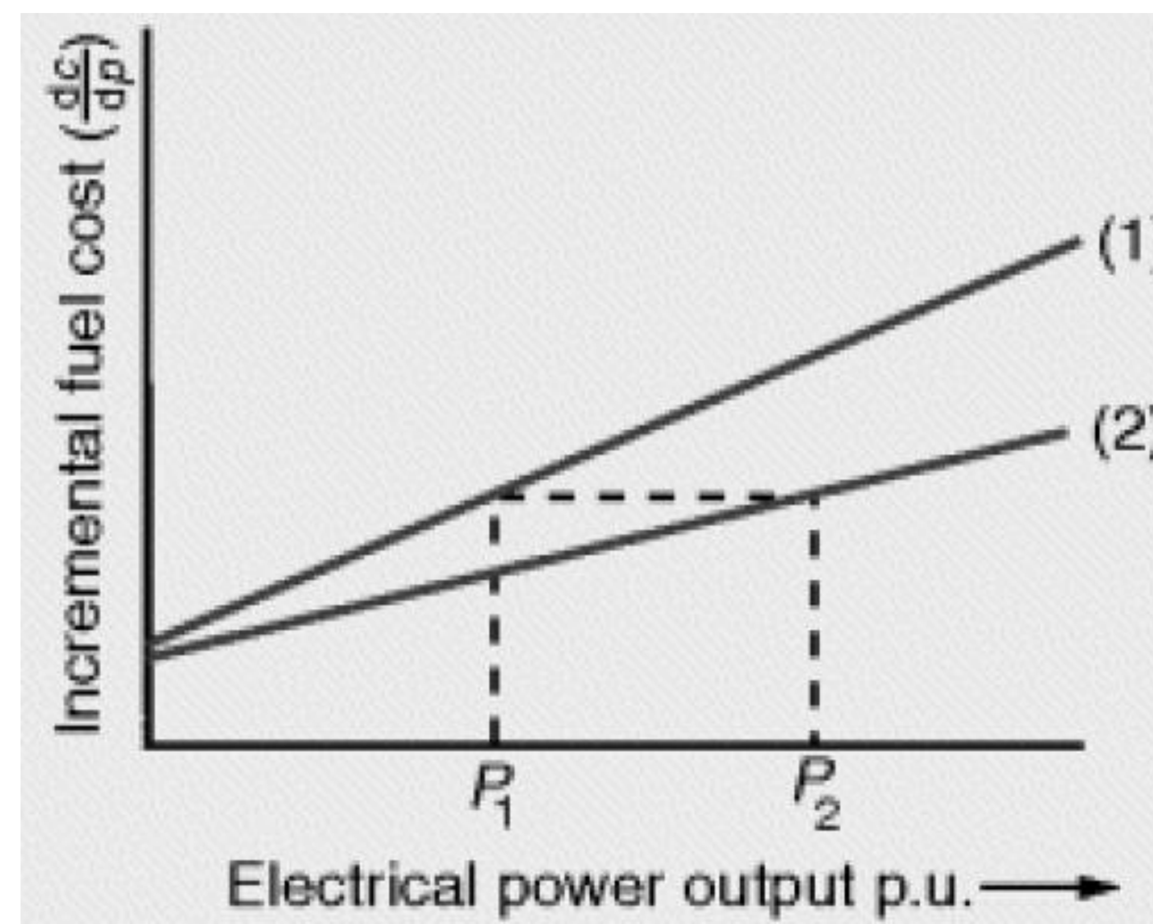


Energy Costs – Calculation Example

Assignment 2:

- What is the incremental fule cost for the coal power plant described in previous example
- Wilans line: $H_{(P)} = 126 + 8.9P + 0.0029P^2$
- Operating point: P: 500 MW
- Fuel cost: Coal 1.8 Euro/MJ





Energy Costs – Calculation Example

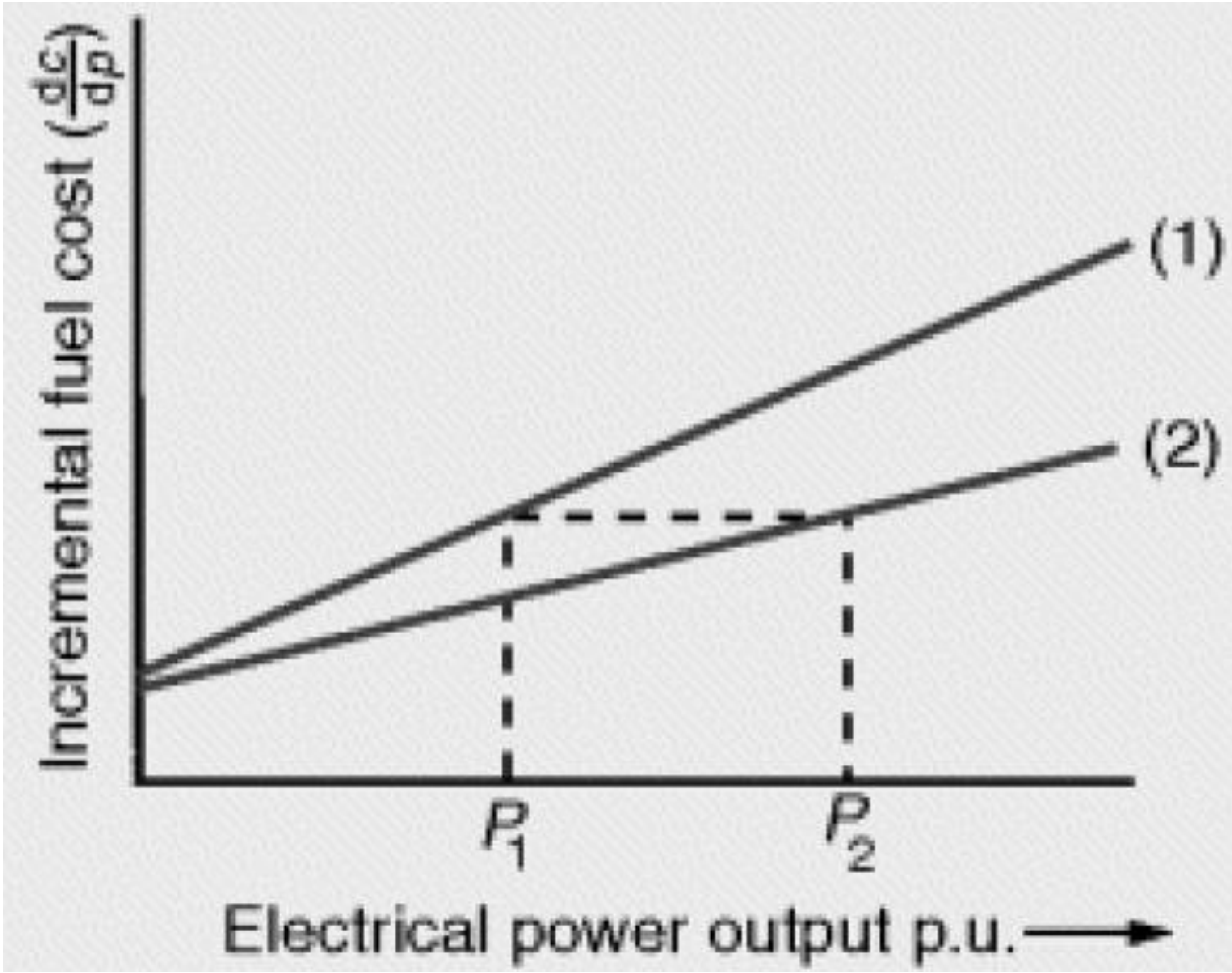
Assignment 3: Find the ideal load sharing between two generators

Generator 1:

- Wilans line: $H1_{(P1)} = 110 + 8.2P1 + 0.002P1^2$
- Minimum stable generation $P1_{Min}$: 100 MW
- Maximum power output $P1_{Max}$: 480 MW

Generator 2:

- Wilans line: $H2_{(P1)} = 170 + 9.6P2 + 0.001P2^2$
- Minimum stable generation $P2_{Min}$: 230 MW
- Maximum power output $P2_{Max}$: 660 MW



Find the operation optimum (P1, P2) for these two generators when the system demand is 750 MW

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